

Using Propensity Scores

Lucy D'Agostino McGowan

Wake Forest University

Propensity scores

Matching

Weighting

Stratification

Direct Adjustment

...



Ingredients

150g unsalted butter, plus extra for greasing
150g plain chocolate, broken into pieces
150g plain flour
½ tsp baking powder
½ tsp bicarbonate of soda
200g light muscovado sugar
2 large eggs

Method

1. Heat the oven to 160C/140C fan/gas 3. Grease and base line a 1 litre heatproof glass pudding basin and a 450g loaf tin with baking parchment.
2. Put the butter and chocolate into a saucepan and melt over a low heat, stirring. When the chocolate has all melted remove from the heat.



estimand

estimator

estimate

Propensity scores

Matching

Weighting

Stratification

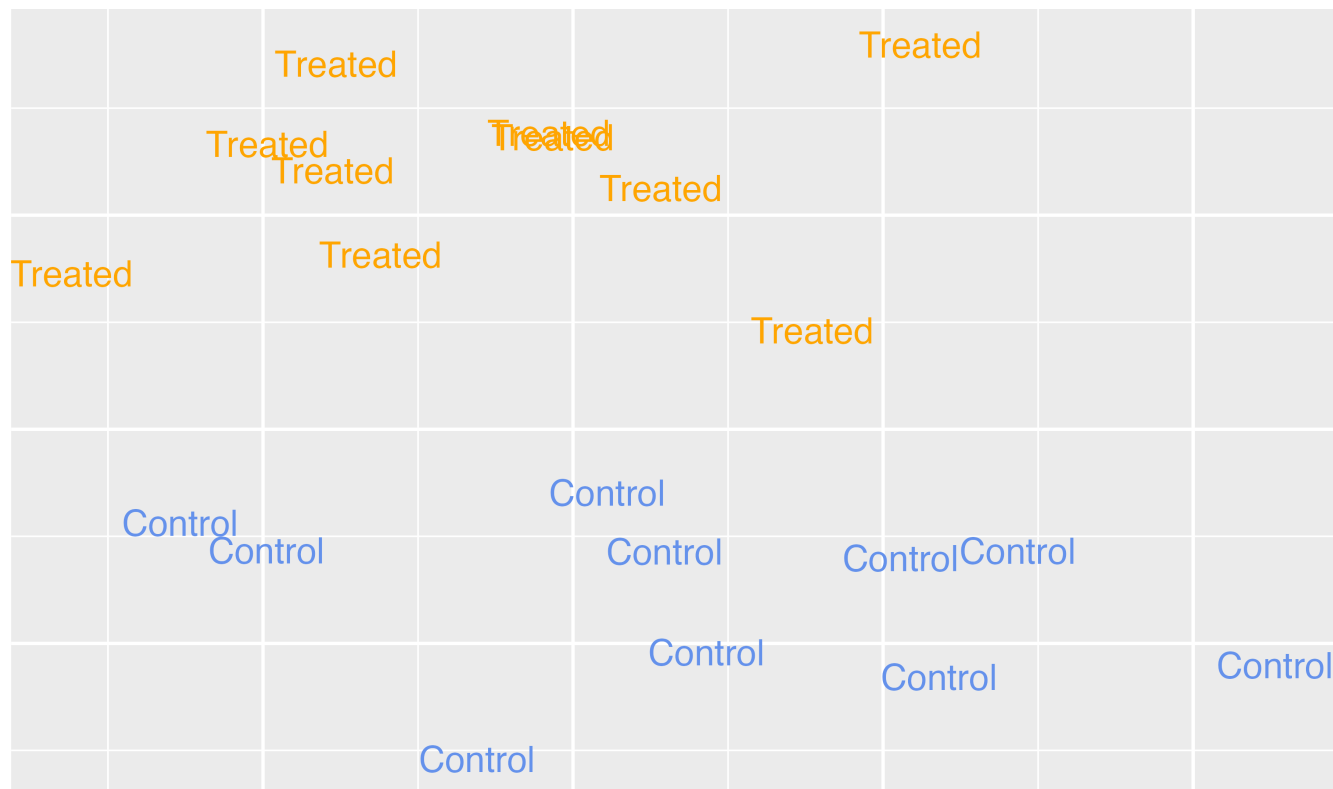
Direct Adjustment

...

Target estimands

Average Treatment Effect (ATE)

$$\tau = E[Y(1) - Y(0)]$$



Target estimands

Estimand	Target population	Example Research Question
ATE	Full population	<i>Should we decide whether to have extra magic hours all mornings to change the wait time for Seven Dwarfs Mine Train between 9-10 AM?</i> <i>Should a specific policy be applied to all eligible observations?</i>

Target estimands

Average Treatment Effect among the Treated (ATT)

$$\tau = E[Y(1) - Y(0) | Z = 1]$$

Target estimands

Estimand	Target population	Example Research Question
ATT	Exposed (treated) observations	<i>Should we stop extra magic hours to change the wait time for Seven Dwarfs Mine Train between 9-10 AMpm?</i> <i>Should we stop our marketing campaign to those currently receiving it?</i> <i>Should medical providers stop recommending treatment for those currently receiving it?</i>

Matching in R (ATT)

```
1 library(MatchIt)
2 m <- matchit(
3   qsmk ~ sex +
4     race + age + I(age^2) + education +
5     smokeintensity + I(smokeintensity^2) +
6     smokeyrs + I(smokeyrs^2) + exercise +
7     active + wt71 + I(wt71^2),
8   data = nhfs_complete
9 )
10 m
```

A 'matchit' object

- method: 1:1 nearest neighbor matching without replacement
- distance: Propensity score
 - estimated with logistic regression
- number of obs.: 1566 (original), 806 (matched)
- target estimand: ATT
- covariates: sex, race, age, I(age^2), education, smokeintensity, I(smokeintensity^2), smokeyrs, I(smokeyrs^2), exercise, active, wt71, I(wt71^2)

Matching in R (ATT)

```
1 matched_data <- get_matches(m, id = "i")
2 as_tibble(matched_data)
```

```
# A tibble: 806 × 71
```

	i	subclass	weights	seqn	qsmk	death	yrdth	modth
	<chr>	<fct>	<dbl>	<dbl>	<dbl>	<dbl>	<dbl>	<dbl>
1	11	1	1	428	1	0	NA	NA
2	1220	1	1	23045	0	0	NA	NA
3	15	2	1	446	1	1	88	1
4	1082	2	1	22294	0	0	NA	NA
5	18	3	1	596	1	0	NA	NA
6	534	3	1	14088	0	0	NA	NA
7	23	4	1	618	1	0	NA	NA
8	697	4	1	18085	0	0	NA	NA
9	27	5	1	806	1	0	NA	NA
10	879	5	1	21128	0	0	NA	NA

```
" - 706 - - - - -
```

Target estimands

*Average Treatment Effect among the Controls
(ATC)*

$$\tau = E[Y(1) - Y(0)|Z = 0]$$

Target estimands

Estimand	Target population	Example Research Question
ATU	Unexposed (control) observations	<i>Should we add extra magic hours for all days to change the wait time for Seven Dwarfs Mine Train between 9-10 AMpm?</i> <i>Should we extend our marketing campaign to those not receiving it?</i> <i>Should medical providers extend treatment to those not currently receiving it?</i>

Matching in R (ATC)

```
1 m <- matchit(  
2   qsmk ~ sex +  
3     race + age + I(age^2) + education +  
4     smokeintensity + I(smokeintensity^2) +  
5     smokeyrs + I(smokeyrs^2) + exercise +  
6     active + wt71 + I(wt71^2),  
7   data = nhefs_complete,  
8   estimand = "ATC"  
9 )  
10 m
```

A `matchit` object

- method: 1:1 nearest neighbor matching without replacement
- distance: Propensity score
 - estimated with logistic regression
- number of obs.: 1566 (original), 806 (matched)
- target estimand: ATC
- covariates: sex, race, age, I(age^2), education, smokeintensity, I(smokeintensity^2), smokeyrs, I(smokeyrs^2), exercise, active, wt71, I(wt71^2)

Target estimands

Average Treatment Effect among the Matched (ATM)

Target estimands

Estimand	Target population	Example Research Question
ATM	Evenly matchable	<i>Are there some days we should change whether we are offering extra magic hours in order to change the wait time for Seven Dwarfs Mine Train between 9-10 AMpm?</i> <i>Is there an effect of the exposure for some observations?</i> <i>Should those at clinical equipoise receive treatment?</i>

Matching in R (ATM)

```
1 m <- matchit(  
2   qsmk ~ sex +  
3     race + age + I(age^2) + education +  
4     smokeintensity + I(smokeintensity^2) +  
5     smokeyrs + I(smokeyrs^2) + exercise +  
6     active + wt71 + I(wt71^2),  
7   data = nhefs_complete,  
8   link = "linear.logit",  
9   caliper = 0.1  
10 )  
11 m
```

Observations with propensity scores (on the linear logit scale) within 0.1 standard errors (the caliper) will be discarded

Matching in R (ATM)

A `'matchit'` object

- method: 1:1 nearest neighbor matching without replacement
- distance: Propensity score [caliper]
 - estimated with logistic regression and linearized
- caliper: `<distance>` (0.063)
- number of obs.: 1566 (original), 780 (matched)
- target estimand: ATT
- covariates: sex, race, age, $I(\text{age}^2)$, education, smokeintensity, $I(\text{smokeintensity}^2)$, smokeyrs, $I(\text{smokeyrs}^2)$, exercise, active, wt71, $I(\text{wt71}^2)$

Matching in R (ATM)

```
1 matched_data <- get_matches(m, id = "i")
2 as_tibble(matched_data)
```

A tibble: 780 × 71

	i	subclass	weights	seqn	qsmk	death	yrdth	modth
	<chr>	<fct>	<dbl>	<dbl>	<dbl>	<dbl>	<dbl>	<dbl>
1	11	1	1	428	1	0	NA	NA
2	1220	1	1	23045	0	0	NA	NA
3	15	2	1	446	1	1	88	1
4	1082	2	1	22294	0	0	NA	NA
5	18	3	1	596	1	0	NA	NA
6	534	3	1	14088	0	0	NA	NA
7	23	4	1	618	1	0	NA	NA
8	697	4	1	18085	0	0	NA	NA
9	27	5	1	806	1	0	NA	NA
10	879	5	1	21128	0	0	NA	NA

" ... 770 ...

Your Turn 1

Using the propensity scores you created in the previous exercise, create a “matched” data set using the ATM method with a caliper of 0.2.

Propensity scores

Matching

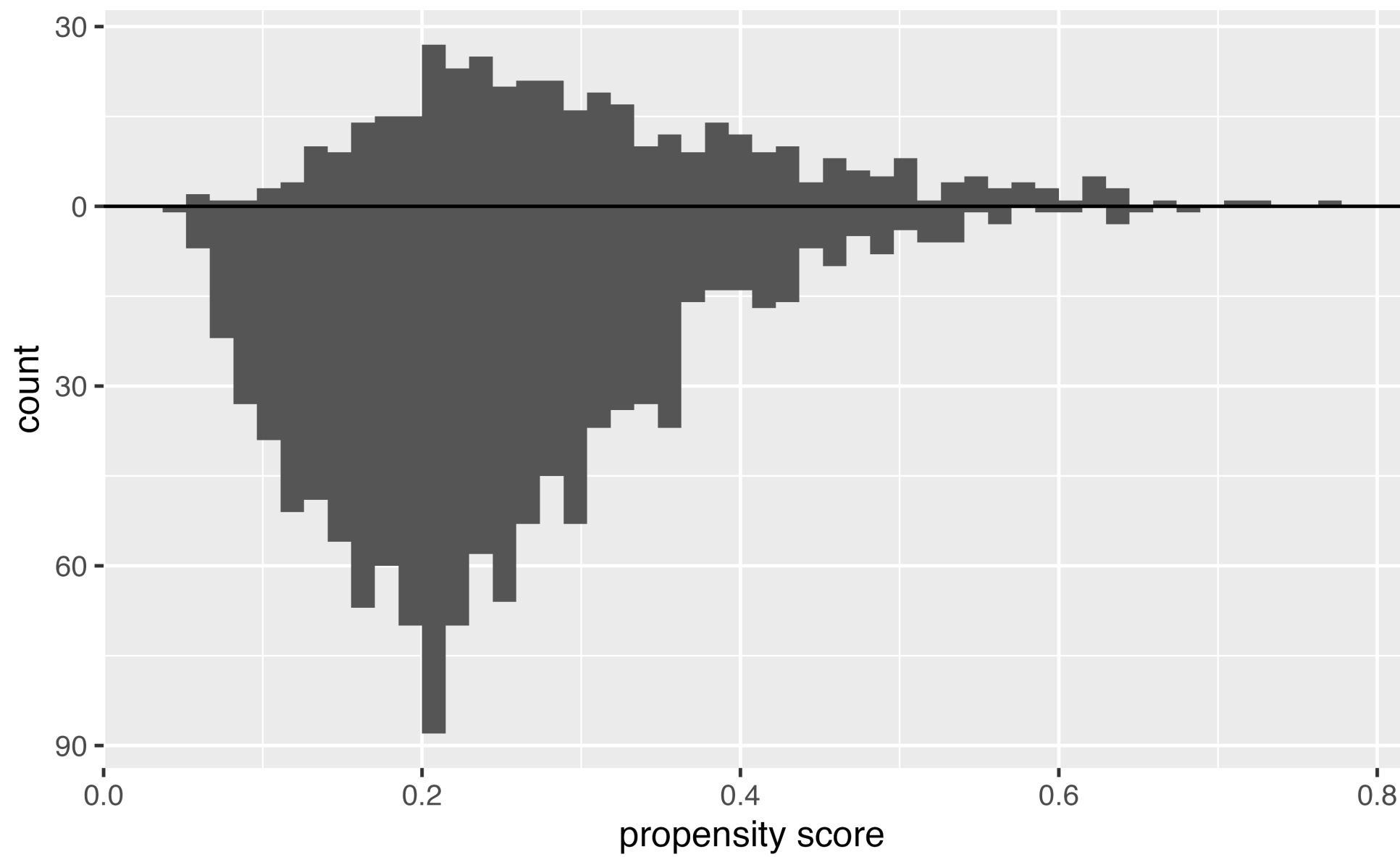
Weighting

Stratification

Direct Adjustment

...

Histogram of propensity scores



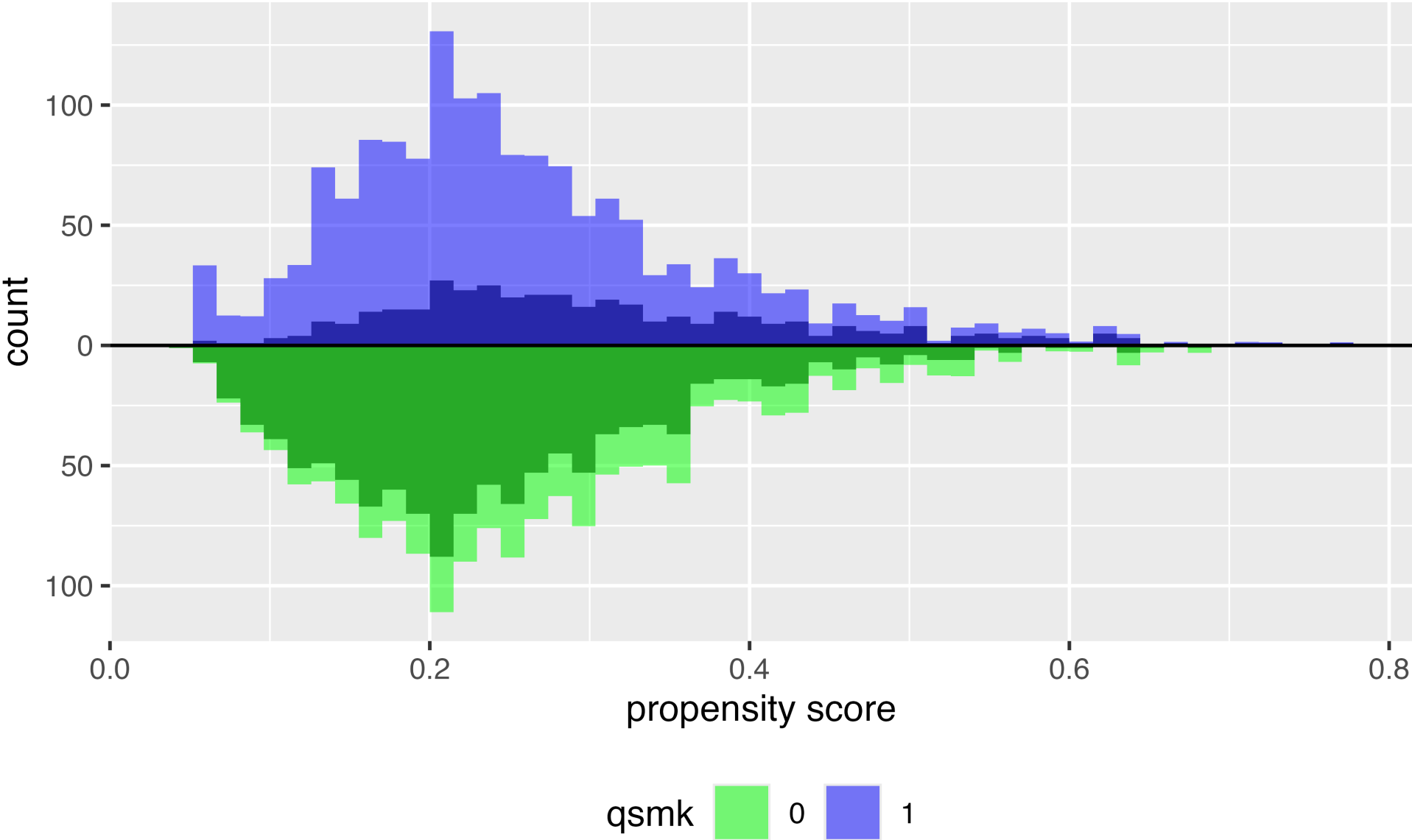
Target estimands: ATE

Average Treatment Effect (ATE)

$$w_{ATE} = \frac{Z_i}{p_i} + \frac{1 - Z_i}{1 - p_i}$$

$$1 \cdot (Z / p) + ((1 - Z) / (1 - p))$$

ATE



Target estimands: ATT & ATC

Average Treatment Effect Among the Treated (ATT)

$$w_{ATT} = \frac{p_i Z_i}{p_i} + \frac{p_i(1 - Z_i)}{1 - p_i}$$

$$1 \quad ((p * Z) / p) + ((p * (1 - Z)) / (1 - p))$$

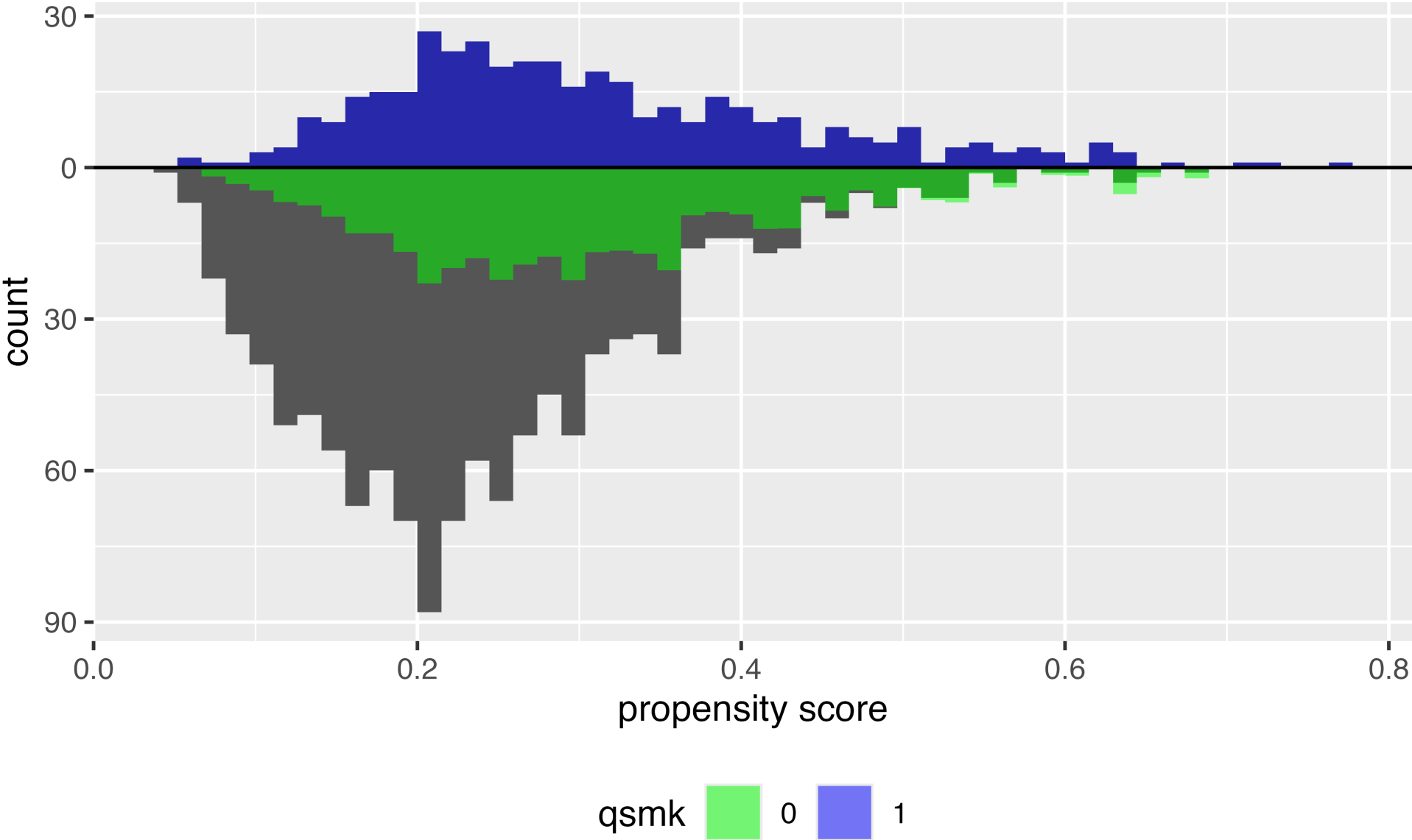
Target estimands: ATT & ATC

Average Treatment Effect Among the Controls (ATC)

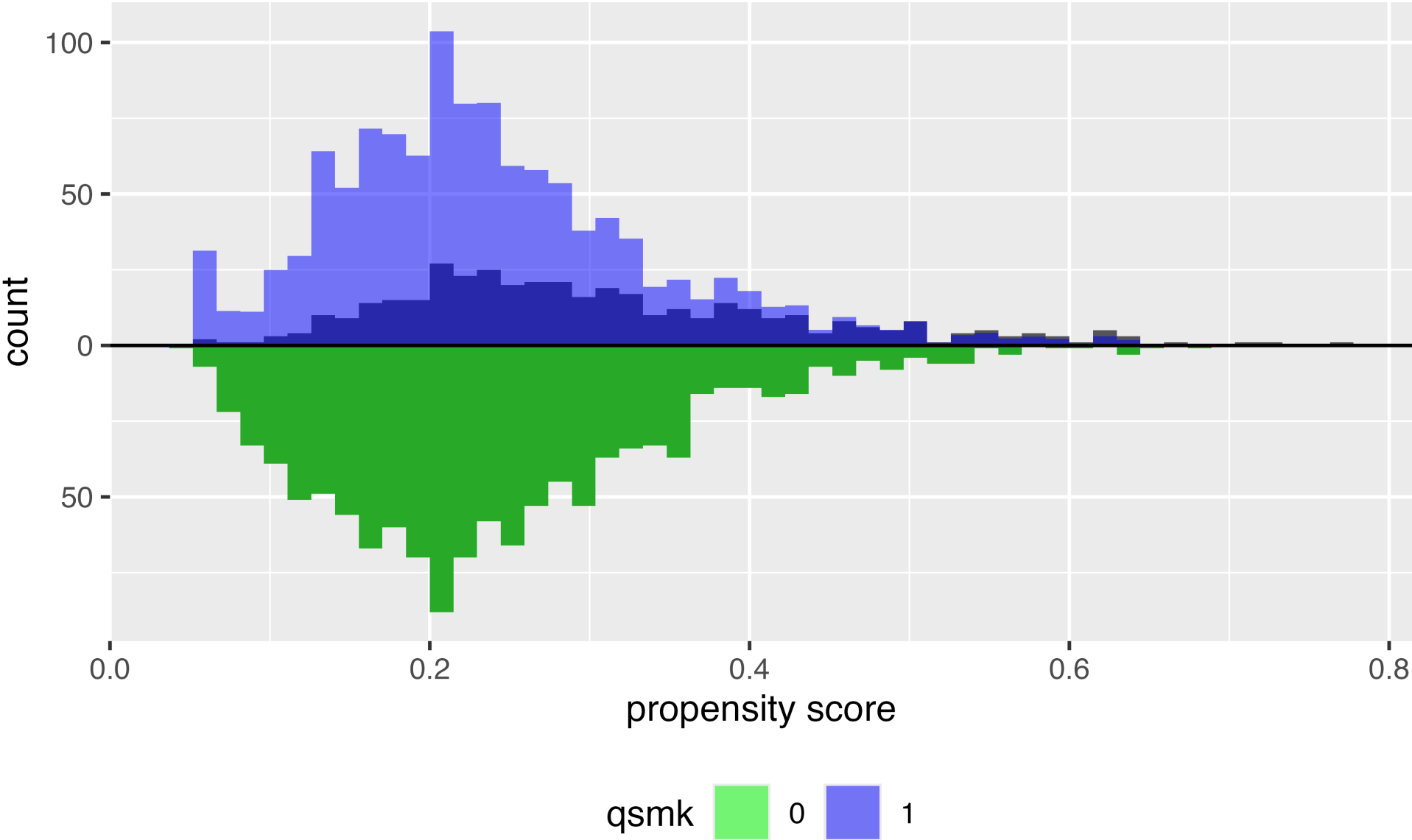
$$w_{ATC} = \frac{(1 - p_i)Z_i}{p_i} + \frac{(1 - p_i)(1 - Z_i)}{(1 - p_i)}$$

$$1 \quad (((1 - p) * Z) / p) + (((1 - p) * (1 - Z)) / (1 - p))$$

ATT



ATC



Target estimands: ATM & ATO

Average Treatment Effect Among the Evenly Matchable (ATM)

$$w_{ATM} = \frac{\min\{p_i, 1 - p_i\}}{Z_i p_i + (1 - Z_i)(1 - p_i)}$$

```
1 pmin(p, 1 - p) / (Z * p + (1 - Z) * (1 - p))
```

Target estimands: ATM & ATO

Average Treatment Effect Among the Overlap Population

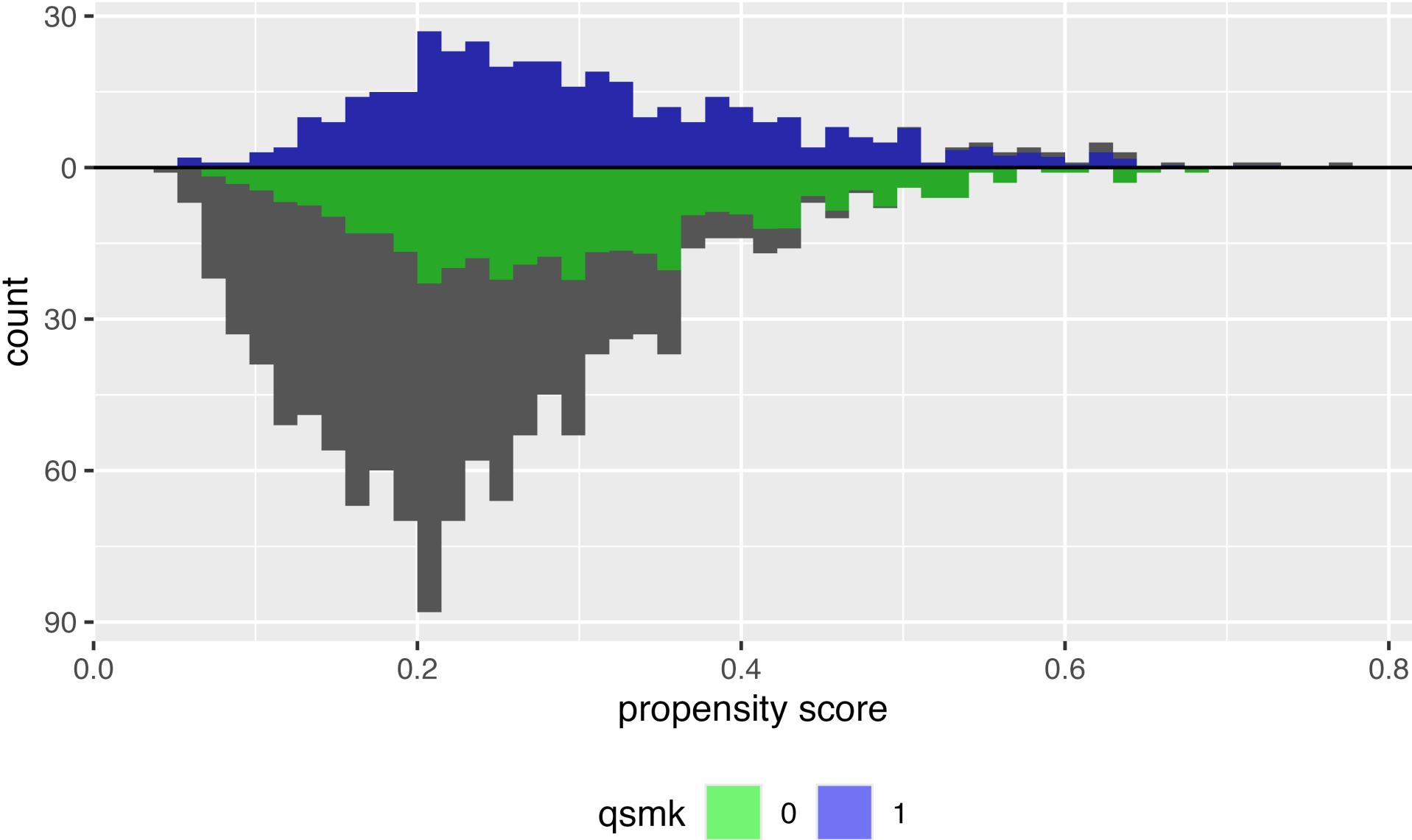
$$w_{ATO} = (1 - p_i)Z_i + p_i(1 - Z_i)$$

$$1 \quad (1 - p) * Z + p * (1 - Z)$$

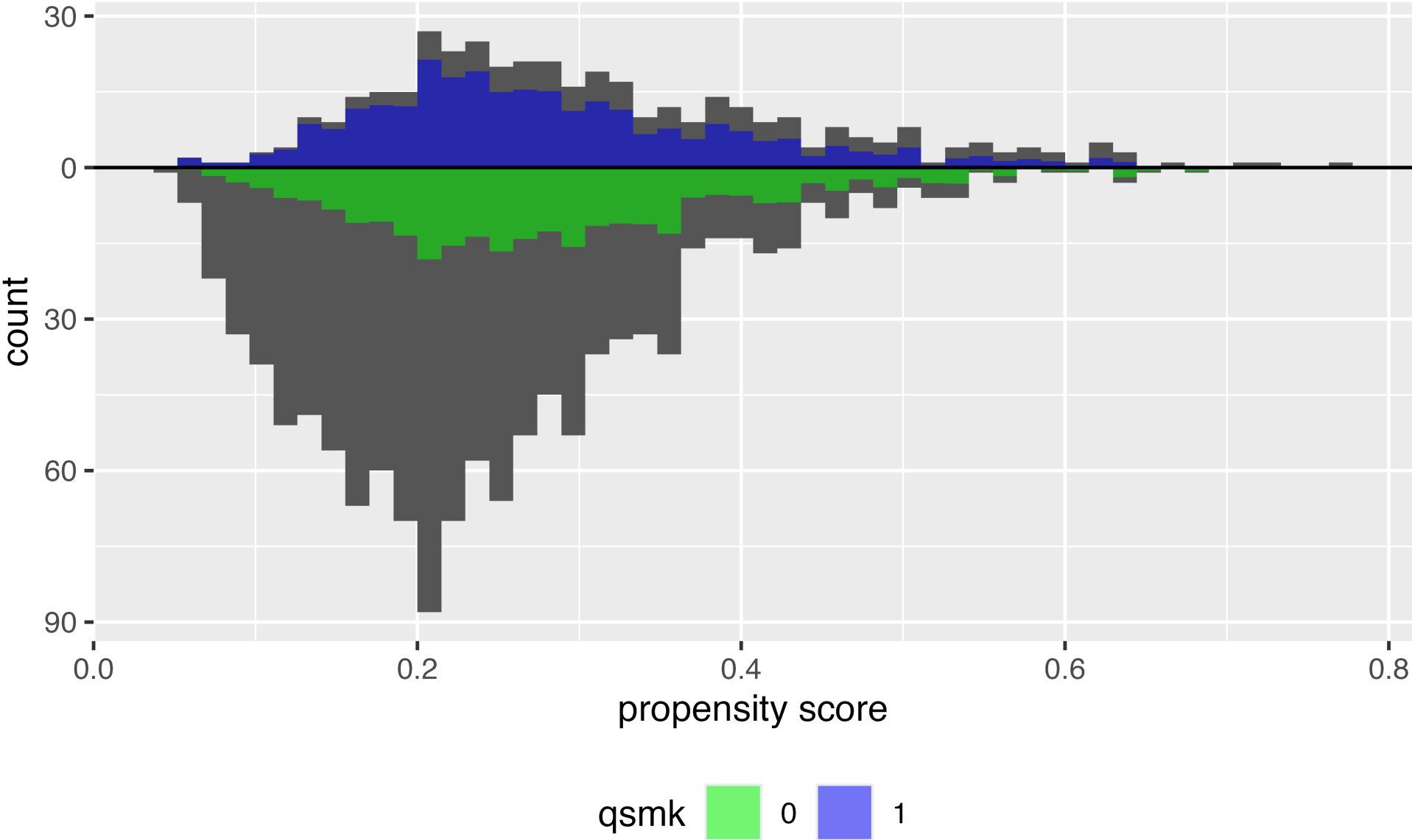
Target estimands

Estimand	Target population	Example Research Question
ATO	Overlap population	<i>Same as ATM</i>

ATM



ATO



ATE in R



Average Treatment Effect (ATE) $w_{ATE} = \frac{Z_i}{p_i} + \frac{1-Z_i}{1-p_i}$

```
1 library(propensity)
2 df <- propensity_model |>
3   augment(type.predict = "response", data = nhefs_complete) |>
4   mutate(w_ate = wt_ate(.fitted, qsmk))
```

Your Turn 2

Using the propensity scores you created in the previous exercise, add the ATE weights to your data frame

Stretch: Using the same propensity scores, create ATM weights